# Electric Charge and Methods of Charging

Questions, Textbook; Answers, Elias Xu

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## 1

*Consider a electrically neutral object. If this object is given a positive charge, does it mean mass increases, decrease, or stay the same? Explain.*
Decreases, because a positive charge means a decrease in the number of electrons, which while they have a small mass, will nonetheless cause mass to decrase.

## 2

*Explain why a comb that has been rubbed through your hair attracts small bits of paper, even though the paper is uncharged?*
Neutral objects, when exposed to a charged object, polarize, leading to them being attracted to the charged comb.

## 3

*A charged rod is brought near a suspended object, which is repelled by the rod. Can we conclude that the suspended object is charged?*
Yes, because if it wasn't charged, the polarized object would attract to the charged rod.

## 4

*A container holds a gas consisting of 1.5 moles of oxygen molecules. One in a million of these molecules has lost a single electron. What is the next charge of the gas?*
To solve, you have the calculate the number of charged particles, and then convert those elementary charges to columbs.

$$\frac{1.5\text{m} \times 6.023 \times 10^2 3 \frac{\text{particles}}{\text{mole}}}{1 \times 10^6 \frac{\text{particles}}{\text{charged particles}}} \times \frac{1.6 \times 10^{-19}\text{C}}{e} = \boxed{0.144C}$$

The first part compute the number of charged particles, and the second part converts those charged particles into columbs. $6.023 \times 10^2 3 \frac{\text{particles}}{\text{mole}}$ is Avogadro's number.

## 5

*When adhesive tape is pulled from a dispenser, the detached tape acquired a positive charge and the remaining tape in the despenser aquires a negative charge. If the tape pulled from the dispenser has* $0.14\mu C$ *of charge per centimeter, what length of tape must be pulled to transfer* $1.8 \times 10^1 3$ *electrons to the remaining tape?*
You just have to divide the amount of charge needed by the rate, after converting the electrons to columbs.

$$1.8 \times 10^1 3 \text{ electrons} \times 1.6 \times 10^{-19} \frac{\text{C}}{\text{electrons}} = 2.88 \times 10^{-6}\text{C}$$

$$\frac{2.88 \times 10^{-6}\text{C}}{0.14 \times 10^{-6} \frac{\text{C}}{\text{cm}}} = \boxed{20.57\text{cm}}$$

## 6

*A system of 1525 particles, each which is either a electron or a proton, has a net charge of* $-5.456 \times 10^{-17} C$ ***(a)*** *How many electrons are in this system?* ***(b)*** *What is the mass of this system?*

To solve this problem, it's better to think about how many **more** electrons there are than protons, given the negative charge. To find that number, one does

$$\frac{-5.456 \times 10^{-17}\mathrm{C}}{-1.6 \times 10^{-19}\frac{\mathrm{C}}{\mathrm{electron}}} = 341 \text{ electrons}$$

And thus, the rest of the 1184 particles are made up equally of electrons and protons, you get the answer $\boxed{933}$ for **(a)**

To solve for **(b)**, one just calculates the mass based on the predetermined number of protons and electrons.

$$\text{mass } = 592 \text{ protons } \times 1.672 \times 10^{-25}\mathrm{kg} \ + 933\text{electrons} \times 9.109 \ times 10^{-31}\mathrm{kg} = 9.906 \times 10^{-25}\mathrm{kg}$$